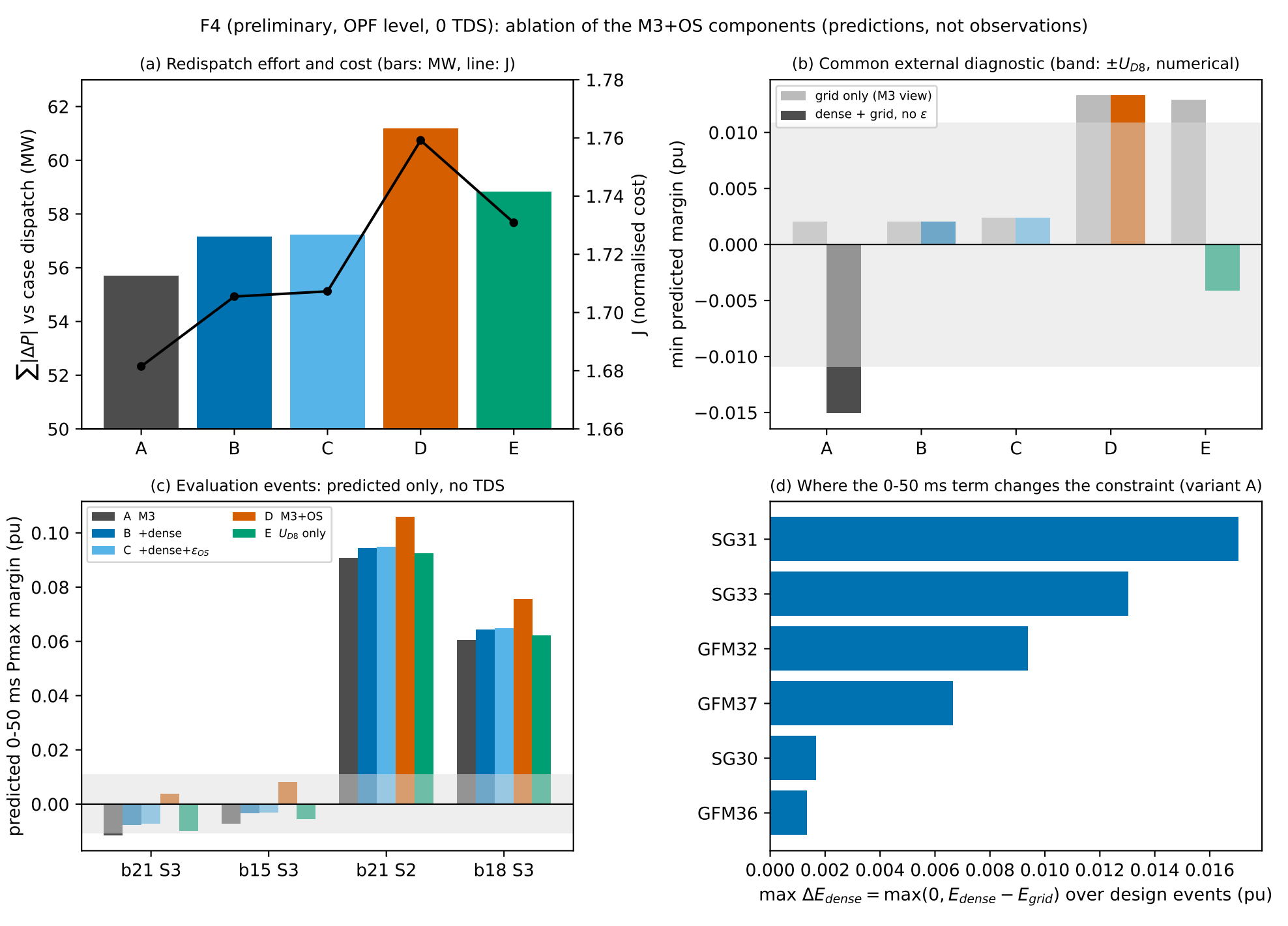
F4 (preliminary, OPF level, 0 TDS): ablation of the M3+OS components (predictions, not observations)
(a) Redispatch effort and cost (bars: MW, line: J)
$\sum|\Delta P|$ vs case dispatch (MW)
J (normalised cost)
A
B
C
D
E
(b) Common external diagnostic (band: $\pm U_{D8}$, numerical)
grid only (M3 view)
dense + grid, no ε
min predicted margin (pu)
(c) Evaluation events: predicted only, no TDS
A M3
B +dense
C +dense+ε_{OS}
D M3+OS
E U_{D8} only
predicted 0-50 ms Pmax margin (pu)
b21 S3
b15 S3
b21 S2
b18 S3
(d) Where the 0-50 ms term changes the constraint (variant A)
SG31
SG33
GFM32
GFM37
SG30
GFM36
max $\Delta E_{dense} = \max(0, E_{dense} - E_{grid})$ over design events (pu)